

Fluid and Cell Behaviors Along a 3D Printed Alginate/Gelatin/Fibrin Channel

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ABSTRACT: Three-dimensional (3D) cell manipulation is available with the integration of microfluidic technology and rapid prototyping techniques. High-Fidelity (Hi-Fi) constructs hold enormous therapeutic potential for organ manufacturing and regenerative medicine. In the present paper we introduced a quasi-three-dimensional (Q3D) model with parallel biocompatible alginate/gelatin/fibrin hurdles. The behaviors of fluids and cells along the microfluidic channels with various widths were studied. Cells inside the newly designed microfluidic channels attached and grew well. Morphological changes of adipose-derived stem cells (ADSCs) in both two-dimensional (2D) and 3D milieu were found on the printed constructs. Endothelialization occurred with the co-cultures of ADSCs and hepatocytes. This study provides insights into the interactions among fluids, cells and biomaterials, the behaviors of fluids and cells along the microfluidic channels, and the applications of Q3D techniques.

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Introduction

Organ donor shortage has been a serious social problem for many years due to heavy traffic accidents and organ failures (Badylak et al., 2012; Rahaman and Mao, 2005; Verhulsel et al., 2014; Wang et al., 2007, 2009; Wang, 2012). Organ manufacturing is a promising approach to solve this problem in the near future (Cui et al., 2009; He and Wang, 2011; Li et al., 2009a,b; Wang and Xu, 2010; Xu et al., 2007, 2008a,b; Yan et al., 2005a,b). At present, this approach is still

in its infancy with several bottlenecks in clinical transplantation and medical application (Wang, 2013; Wang and Zhang, 2011; Wang et al., 2010, 2013). One of the bottlenecks is to rapidly build a vascular system with essential cell types in an organized construct.

During the last decade, both fluidic manipulation and rapid prototyping (RP), also known as additive manufacturing (AM) or three-dimensional printing (3DP), technologies have witnessed their rapid development stages with great achievements. For example, various 3D bio-constructs have been produced which can greatly mimic cell aggregations in native tissues/organs for the study of cell-biomaterial interaction, cell-cell signaling modulation, and multiple cell co-culture effect (Lin and Anseth, 2011). Some elaborate micro/macro conduits/tubes have been fabricated which can act as ideal hosts for biochemical reactions (Bhatia and Ingber, 2014; Duan et al., 2014; Tibbitt and Anseth, 2012). Diversified fluidic reactors with cell-laden or cell-seedable hydrogels have been created which can be customized with optimum microstructures and cell viabilities (Choi et al., 2007; Cushing and Anseth, 2007). Several nano- or micro-size fluidic devices have been developed to enable scientists to manipulate single cells, organelles and even macromolecules, including DNA and RNA, automatically and repetitively (Horrocks et al., 2013; Northrup, 2004; Wang and Li, 2011). With these fluidic manipulation technologies, scientists can perform a large spectrum of cellular and chemical arrays for shear force characterization, chemical reaction analysis, cell/drug screening, protein crystallization, etc.

In parallel, 3DP constructs with categories of biomaterials and go-through channels have been frequently reported. For example, numerous grid polymeric scaffolds have been presented for cytological and histological studies (Bellan et al., 2009; Flachsbarth et al., 2006; Reed et al., 2001). Thousands of quasi-three-dimensional (Q3D) structures have been demonstrated using liquid (oil and water) with droplet writing techniques (Derby, 2010, 2012; Khoshmanesh et al., 2011; Kitson et al., 2012; Lee et al., 2009; Villar et al., 2013). A few micro channels have been printed using sacrificial 3D carbohydrate (Miller et al., 2012). Especially, a series of 3DP (RP or AM) technologies using both synthetic polymers, such as poly(lactic-co-glycolic acid (PLGA) and polyurethane (PU), and

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natural polymers, such as gelatin-based hydrogels, have been investigated in our own group for complex organ manufacturing (Wang, 2014; Wang et al., 2006; Xu et al., 2010). Nevertheless, nearly all the existing fluidic and 3DP techniques have been separated obviously without providing deep insights into the interactions among fluids, cells and biomaterials.

It was hypothesized that intricately patterned and functionally compartmentalized bionic structures with alterable channels and tactic biomaterials can be obtained by the integration of flow fields, living cell manipulations and intelligent freeform 3DP technologies (Huh et al., 2011; Kolewe et al., 2013; Melchels et al., 2012). Meanwhile, biological factors and physicochemical environments could greatly affect cell activities, particularly stem cells behaviors (Hunt, 2008). With regard to stem cell techniques or therapeutics, two-dimensional (2D) or 3D microfluidic structures could help to study the natural signaling factors or synthetic small molecules in tissue/organ regeneration (Ertl et al., 2014; Ginebra et al., 2010; Hasan et al., 2014; Huang et al., 2012; Oh et al., 2009). Furthermore, the combination of microfluidic actions, living cell manipulation techniques, and intelligent freeform 3DP technologies could provide microenvironment niches for basic probes of fluids and cells in a precisely controllable way (Capel et al., 2013; Wilson and McDevitt, 2013).

In the present paper, we fabricated a Q3D construct with parallel alginate/gelatin/fibrin hydrogel hurdles with our own developed RP device (Wang et al., 2006, 2013; Xu et al., 2010). The alginate/gelatin/fibrin hurdles acted as an extracellular matrix (ECM) to allow fluid to flow along the relatively-open micro channels in between and to promote cell attachment and growth. The good mechanical performance of the crosslinked/polymerized alginate/fibrin molecules prevents the alginate/gelatin/fibrin hurdles from collapse with the fluid flow. Compared with our previous studies, the microfluidic channels are more open and present larger interfaces for fluid cycling and nutrient exchange. Hepatocytes and adipose-derived stem cells (ADSCs) were applied onto the hurdles with microfluidic flow movements, creating milieus for divisional liver lobular analogs with controllable vascularization processes (Novosel et al., 2011; Qiu et al., 2012; Ricci-Vitiani et al., 2010; Takebe et al., 2013; Yao et al., 2009). Computational simulations were performed on the microfluidic dynamics as a comparison. With the integration of the predesigned 3D printed biocompatible alginate/gelatin/fibrin hurdles and microfluidic techniques, biomimetic milieus were created with divisional zones mimicking vascularized hepatic structures. This integration made it possible to investigate the pertinent fundamental mechanisms on cell viability, cell-material, and cell-cell interactions for regenerative medicine in equilibrium fluidic conditions (Cushing and Anseth, 2007; Fairbanks et al., 2011; Haraldsson et al., 2006; Rydholm et al., 2006; Tibbitt and Anseth, 2009).

Materials and Methods

Cell Culture

ADSCs were isolated from rat subcutaneous adipose tissues as reported [Xu et al., 2009; Xu et al., 2010]. The ADSCs were then cultured in Dulbecco's modified Eagle's medium (DMEM) (Gibco, Santa Cruz, CA) containing 10% fetal bovine serum (FBS, senior tier, Sijiqing, China), 200 kU/L penicillin and 100 mg/L strepto-

mycin in an incubator at 37°C with 5% CO₂. The medium was changed every second day. Cells were passaged three times by standard protocols of trypsinization before use.

Hepatocytes were extracted from rat livers. Hepatocytes of SD rats were infused with 0.25% type II collagenase (Sigma, St. Louis, MO), minced and incubated at 37°C with 5% CO₂ for 5 min. The neutralized cells were then washed in DMEM, centrifuged, and cultured in DMEM containing 10% FBS, 200 kU/L penicillin and 100 mg/L streptomycin in an incubator at 37°C with 5% CO₂. The medium was changed every day.

Fabrication of Gelatin/Alginate/Fibrin Hurdles

Alginate sodium (Sinopharm Chemical Reagent Co., Ltd, SCRC, China) and gelatin (Sigma) were dissolved in phosphate buffered saline (PBS) respectively to form a homogeneous solution with a concentration of 5% or 20% (w/v, polymer weight/solvent volume). The 5% alginate and 20% gelatin solutions were then mixed at a ratio of 1:1 v/v before being pasteurized three times at 70°C for 20 min. Finally, fibrinogen powder was mixed with a sterile solution to form an alginate/gelatin/fibrinogen mixture.

The alginate/gelatin/fibrinogen mixture was loaded into a syringe (10 mL) with a stainless nozzle (0.5 mm in diameter) before being printed [Wang et al., 2006]. In the formation chamber, coverslips and petri dishes made of glasses were put on a piece of ice. The alginate/gelatin/fibrinogen hurdles was then fabricated on the coverslips based on a digital model. The velocity for *X* and *Y* axes was 10 mm/s. Meantime the acceleration was 10 mm/s². After the alginate/gelatin/fibrinogen construct was crosslinked in a 2% (w/v) CaCl₂ solution followed by polymerized in a 100 U/mL thrombin solution, parallel alginate/gelatin/fibrin hydrogel hurdles were finally acquired.

Fluid Flow Behavior With Fluorochromes

A buffered solution containing 0.9% (w/v) NaCl, 0.5% (w/v) CaCl₂, and 10% (w/v) FBS was used as a carrier for fluorochromes. Fluorescein (SCRC, China) and rhodamine B (SCRC, China) were dissolved in the buffered solution separately until saturation before use. In the buffered solution, the 0.9% NaCl was used to increase the osmotic pressure; the 10% FBS was used to provide nutrients for cell growth; the 0.5% CaCl₂ was used to maintain the rigidity of the alginate/gelatin/fibrin hydrogel hurdles.

The fluorochrome containing buffered solutions were added dropwisely along the channels from one corner by an intravenous drip with a stainless nozzle (0.3 mm in diameter). Fluorescence images were obtained under a laser scanning confocal microscope (LSCM, LSM710, Zeiss, Germany). Image analyses including laser intensity distribution were carried out in ZEN packages and Microsoft softwares.

Fluid Flow Behavior With Cell Suspensions

The isolated and purified cells were suspended in DMEM homogeneously before use. The densities of the ADSCs and hepatocytes were 3×10^5 cells/mL and 3×10^7 cells/mL, respectively. Cell suspensions were added dropwisely along the channels

from one corner by an intravenous drip with a stainless nozzle (0.5 mm in diameter). A phase-contrast microscope (BX51, Olympus, Japan) was used to observe the alginate/gelatin/fibrin hurdles and cells, momentarily. The flow behaviors of the ADSCs and hepatocytes along the channels were videoed using a LSCM as described above. Mobility of the cells was recorded in time lapse imaging with the interval time of 500 ms. Ten continuous frames of the video were marked with various colors and finally overlapped.

Immunostaining Analyses

The alginate/gelatin/fibrin hurdles with ADSCs were first washed with PBS three times before being incubated with a primary antibody (rabbit anti CD31; Boster Biotechnology Co Ltd, China) for 10 h at 4°C. Then the samples with ADSCs were incubated with a secondary antibody (FITC conjugated goat anti rabbit; Bmassay, China) and 4',6-diamidino-2-phenylindole (DAPI; Bmassay, China) for 1 h at 37°C. Finally, the constructs were washed in PBS three times.

The alginate/gelatin/fibrin hurdles with hepatocytes were first washed with PBS three times. Hepatocytes on the samples were stained with carboxyfluorescein succinimidyl amino ester (CFSE; AAT Bioquest, Sunnyvale, CA) in advance. Then nuclei of the hepatocytes were incubated with DAPI for 30 min. Finally, the constructs were washed with PBS three times again.

After the ADSCs and hepatocytes were co-cultured, the constructs were incubated with a primary antibody (rabbit anti human Factor VIII; Boster Biotechnology Co Ltd, China) for 10 h at 37°C. Then, the samples were incubated with a secondary antibody (TRITC conjugated goat anti rabbit; Bmassay, China) for 2 h at 37°C. Finally, the samples were incubated with DAPI for 30 min before being washed in PBS three times. The samples were observed with a LSCM as described above. Image acquisition was performed with inverted Zeiss microscope and analyzed with ZEN 2010 physiological packages.

Scanning Electron Microscopy Observation

Samples with cells were fixed with 2.5% glutaraldehyde and then washed in PBS three times. The samples were dehydrated in ethanol with five gradient concentrations (i.e., 50%, 70%, 80%, 90%, and 100%). The dried samples were then soaked in ethanol with 50% tert-butanol. Finally, the samples were immersed in pure tert-butanol before freeze-drying at -20°C. Once the organic solvents were removed, the samples were gold-sputtered for 60 s and observed in a scanning electron microscope (SEM; Quanta200, FEI, Czech).

Computer Simulation of the Fluid Flow

Simulation of the fluid flow was conducted with finite element software ANSYS 15.0 (ANSYS, Pittsburgh, PA), and the FLUENT unit was used to investigate the velocity and flow behavior in the microfluidic channels with a range of channel widths (i.e. 4 mm, 2 mm, 1 mm and 0.5 mm). Geometric modelling was conducted in mesh component systems in ANSYS before fine mesh generation. The meshed fluid solid geometry was then defined as a k-epsilon

water-liquid model with an inlet velocity of 3 m/s at 500 iterations. The residual curves show that calculations of the flow field were convergent. Eventually, the computational fluid dynamics-post (CFD-Post) component was used to show the 3D contours and vectors of the fluid distribution. The relationship between flow velocity and geometrical location could be seen in the post-treated colored contour maps.

Results

3D Printed Parallel Alginate/Gelatin/Fibrin Hurdles

A schematic diagram of the fluid and cell behaviors along 3D printed alginate/gelatin/fibrin hurdles is shown in Figure 1. The 3DP process of the parallel alginate/collagen/fibrin hurdles was conducted at ambient temperature (Fig. 2). Different digital parameters could be applied to the processes via the numerical control system. Microfluidic channels with various widths were consequently fabricated. High accuracy and repeatable parallel alginate/collagen/fibrin hurdles were obtained.

Figure 2C shows the locations of the parallel alginate/collagen/fibrin hurdles under optical microscope. The explicit boundaries between the alginate/collagen/fibrin hurdles and micro channels indicate that the alginate/gelatin/fibrinogen hydrogel was cross-linked/polymerized by the CaCl₂ and thrombin solutions. In the alginate/gelatin/fibrinogen hurdles the alginate molecules were crosslinked by the CaCl₂ solution while the fibrinogen molecules were polymerized by the thrombin solution. The parallel alginate/collagen/fibrin hurdles could be well-preserved in DMEM at 37°C for more than two weeks with no fluid flow disruption. When the fluid was added vertically from one corner (Fig. 2B), it flowed through the whole construct along the channels before flowing out at the diagonal corner. The fluidic flows including PBS (Fig. 2D and E), DMEM and

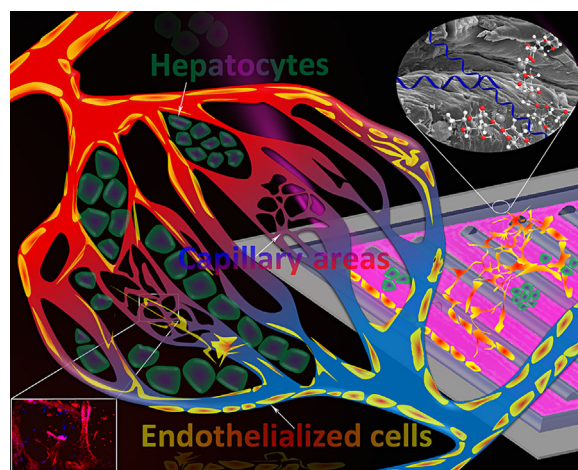


Figure 1. Schematic of the fluid and cell behaviors along a 3D printed alginate/gelatin/fibrin channel. Adipose-derived stem cells (ADSCs) were endothelialized when co-cultured with hepatocytes along the alginate/gelatin/fibrin hurdles and formed hepatic sinusoid like structures.

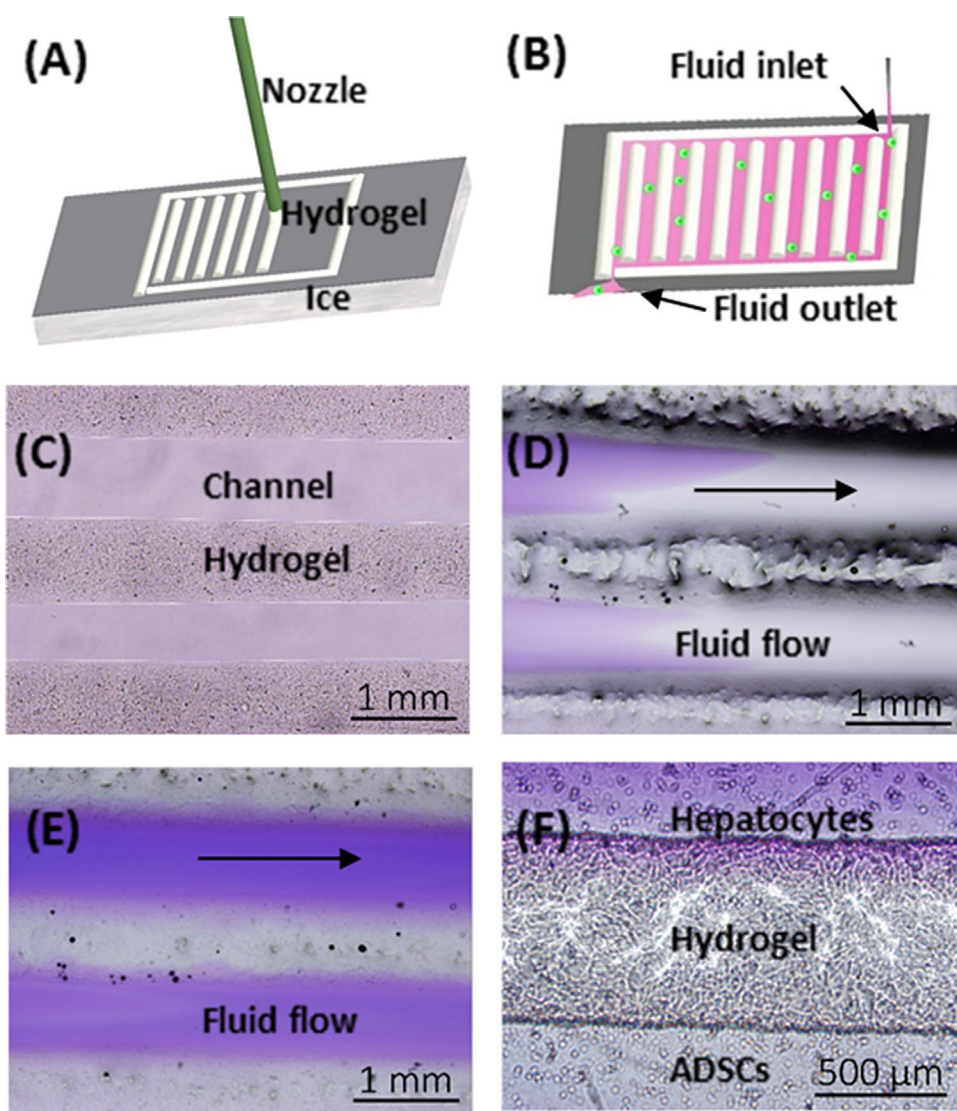


Figure 2. Illustration of the parallel micro channels in between the alginate/gelatin/fibrin hurdles: (A) 3D printing process; (B) schematic of fluid flow with cells; The parallel hydrogel hurdles with fluid and cells: (C) parallel hydrogel hurdles and microfluidic channels (500 μm in width); (D) & (E) PBS flow containing toluidine blue along the channels. The flow direction was marked with an arrow, and the turbulence of the flow could be seen vaguely; (F) hepatocytes and ADSCs suspended in PBS along the channels.

cell suspension (Fig. 2F) along the alginate/collagen/fibrin hurdles were clearly observed with the LSCM.

Fluid Distribution in the Micro Channels

To assess the fluid states along the micro channels, PBS, dyes, as well as cell suspensions were used for microscopy observation. Figure 2D and E show the PBS containing toluidine blue flows along the gelatin/alginate/fibrin hurdles. Discrete cell distribution was found with the ADSC and hepatocyte suspensions (Fig. 2F). The distribution of fluorochromes in liquid droplets and columns under a LSCM are shown in Figure 3. Laser fluorescence intensity analysis shows that the crests of rhodamine B molecules were located at the boundaries between the gelatin/alginate/fibrin hurdles and micro channels (Fig. 3A). In contrast, most of the fluorescein molecules

were distributed in the middle of the channels with no obvious fluctuations (Fig. 3B). The two fluid columns also moved in opposite before they finally met at one interchange point, where substance mutual dissolution occurred afterwards. The distribution intensities of both the fluorescein and rhodamine B eventually become synchronous and homogenous either in parallel or perpendicular direction along the gelatin/alginate/fibrin hurdles (Fig. 3C and D).

Cell Behaviors Over the Alginate/Gelatin/Fibrin Hurdles

Both phase contrast microscope and LSCM were used to observe cell distribution and states in the micro channels (Fig. 2-6). Different cell states were found with different cell suspensions at different time points. ADSC and hepatocyte suspensions were added

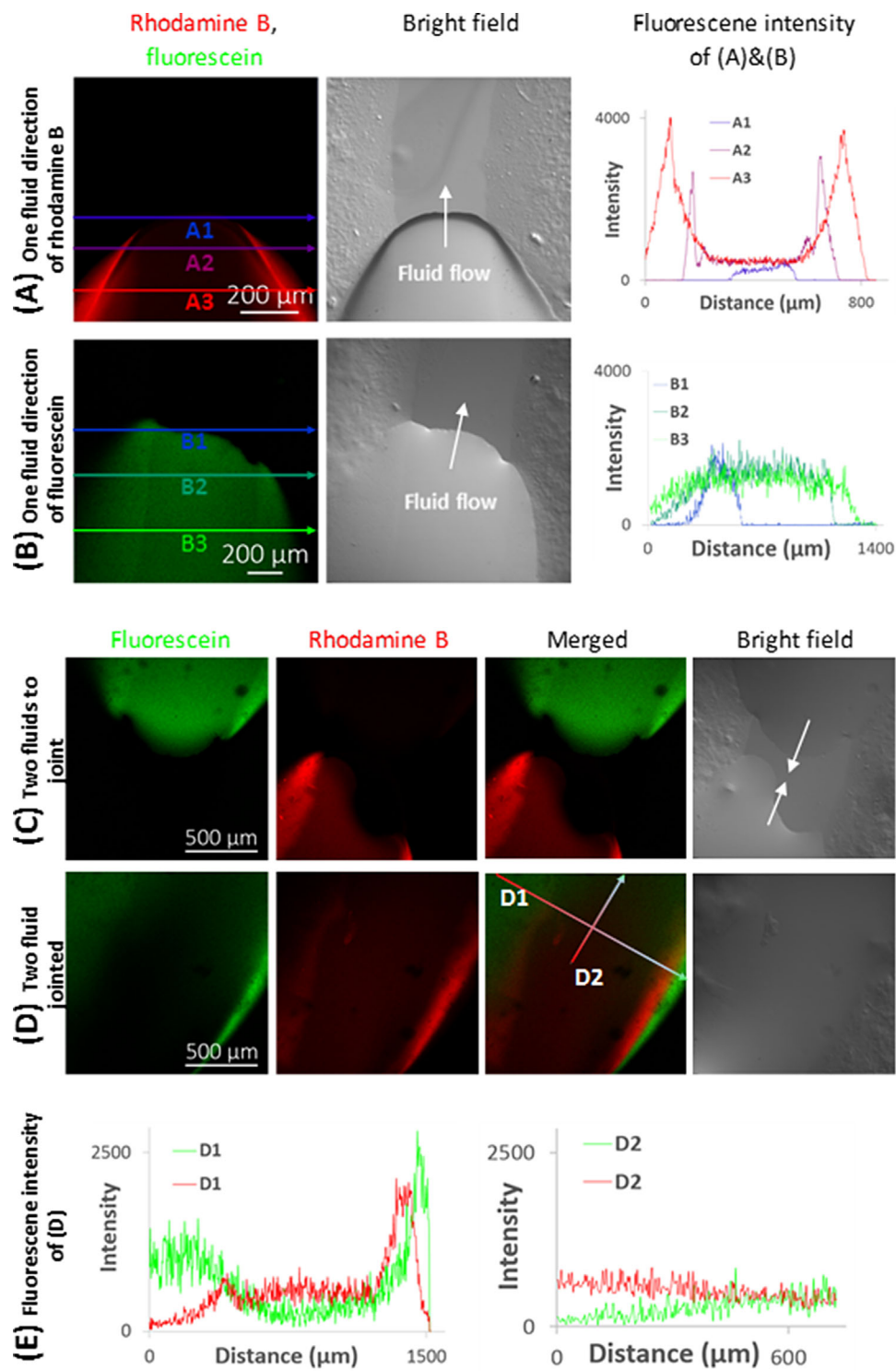


Figure 3. Fluorochrome distribution along the alginate/gelatin/fibrin hurdles: (A) at the beginning of the experiment, one fluid direction of the rhodamine B in a buffered solution; (B) at the beginning of the experiment, one fluid direction of the fluorescein in a buffered solution; (C) & (D) a few seconds later, the interchange area of the two fluids with different fluorochromes.

separately into the Q3D construct from one corner (Fig. 2B). ADSCs located in one channel while hepatocytes concentrated in the adjacent channel. Hepatocyte aggregates were more prominent compared with the scattered ADSCs at the beginning of fluid movement (Fig. 2F). To further demonstrate the cell mobility in the micro channels, cell nuclei were visualized by fluorescence staining and observed dynamically under a LSCM. Figure 4A and B show

that both the ADSCs and hepatocytes distributed uniformly along the micro channels at the beginning of the experiments. Most of the cells moved in lines with the liquid flow along the micro channels. With time goes on, some ADSCs, particularly the ADSC clusters, remained immobile. It was interesting that some cells even moved in direction reversal to the fluid flow, indicating slight turbulence of the fluid.

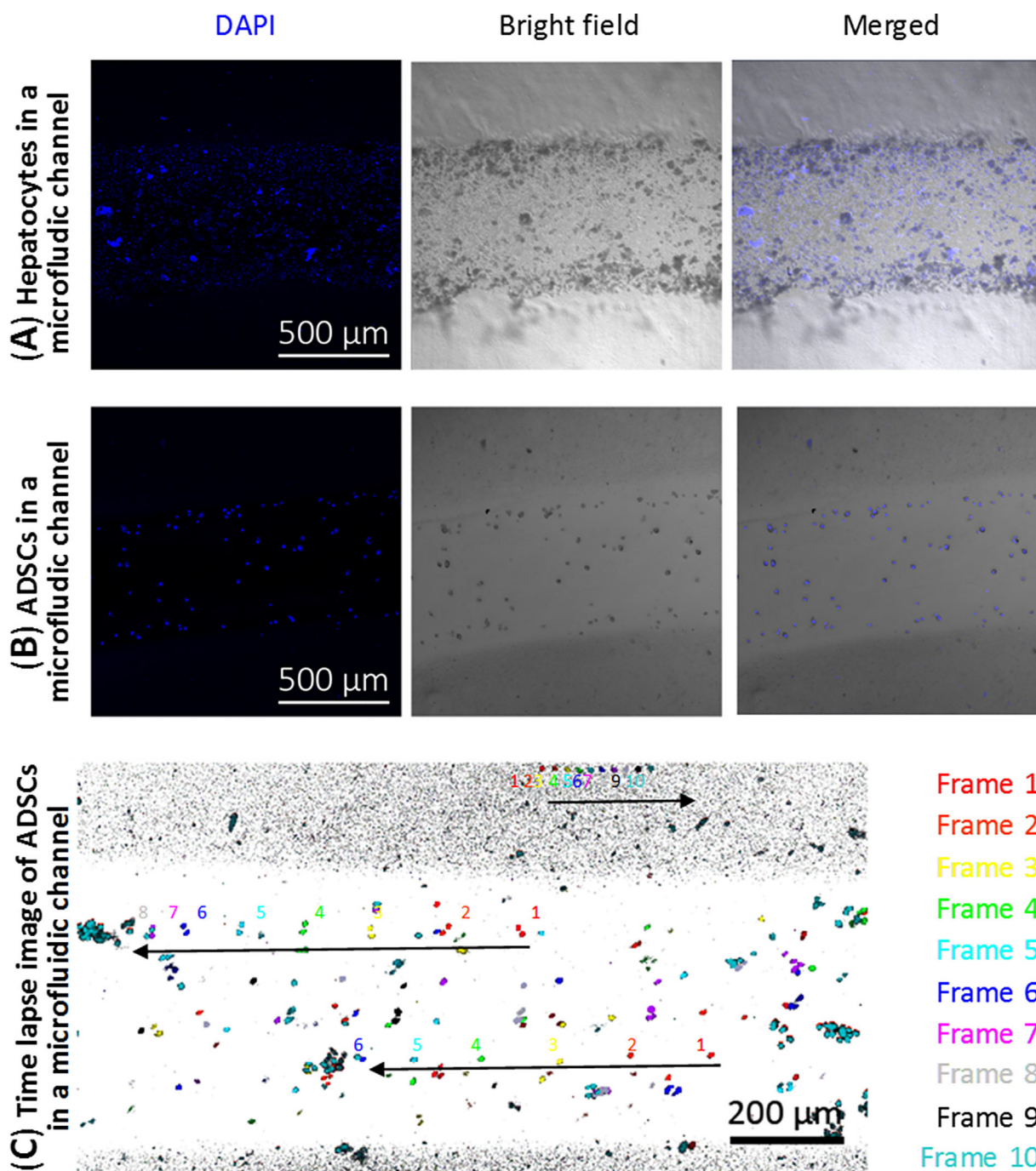


Figure 4. Flow states of ADSCs and hepatocytes in microfluidic channels: (A) flow states of the hepatocytes stained with DAPI; (B) flow states of the ADSCs stained with DAPI; (C) time lapse images of the ADSC flow states stained with DAPI. Images and videos were taken under a LSCM.

Figure 5 shows morphologies of the ADSCs and hepatocytes along/on the alginate/gelatin/fibrin hurdles incubated in DMEM after 10 h and 20 h, respectively. The ADSCs and hepatocytes were separately irrigated through the parallel hurdles. After 10 h of culture, the alginate/gelatin/fibrin hurdles remained vividly distinct (Fig. 5A–D). Interactions of the ADSCs and alginate/gelatin/fibrin hurdles were detected. The ADSCs produced fiber-like pseudopod structures both on the alginate/gelatin/fibrin hydrogel and in the micro channels. The interconnected ADSC networks bridged the neighbor alginate/gelatin/fibrin hurdles and even remote micro channels. Star-like ADSC aggregates on/along the hydrogel hurdles were distinguished. The size of the ADSC networks on the alginate/gelatin/fibrin hurdles was much larger than that in the micro channels. More importantly, most of the boundaries between the alginate/gelatin/fibrin hurdles and the micro channels were covered with linearly arrayed and special patterned ADSC aggregates. After 20 h of culture (Fig. 5E and F), the scales of

the ADSC networks became larger, traversing several hydrogel hurdles and micro channels. In Figure 5G the star-shaped ADSCs spread out along the hydrogel hurdles, while some of the hepatocyte spheres located at the edge of hydrogel hurdles (Fig. 5H). The surface of the alginate/gelatin/fibrin hydrogel was still smooth, which was ideal for endothelial-like cell adherence and migration. After 3–4 days of culture the boundaries of the hydrogel hurdles were jagged slightly, indicating the gradual changes of the thickness of the hydrogel layer. Under a SEM, the Q3D construct shifted and showed a trend of collapse. The alginate/gelatin/fibrin hurdles became thinner, indicating some polymers, such as un-crosslinked gelatin molecules, had permeated the hurdles.

The cellular morphological changes were verified using immune staining under a LSM (Fig. 6). ADSCs tended to aggregate in the Q3D construct over time, even with the fluid flow. ADSCs formed clusters on the border between the alginate/gelatin/fibrin hurdles and micro channels (Fig. 6A and B), which are consistent with the

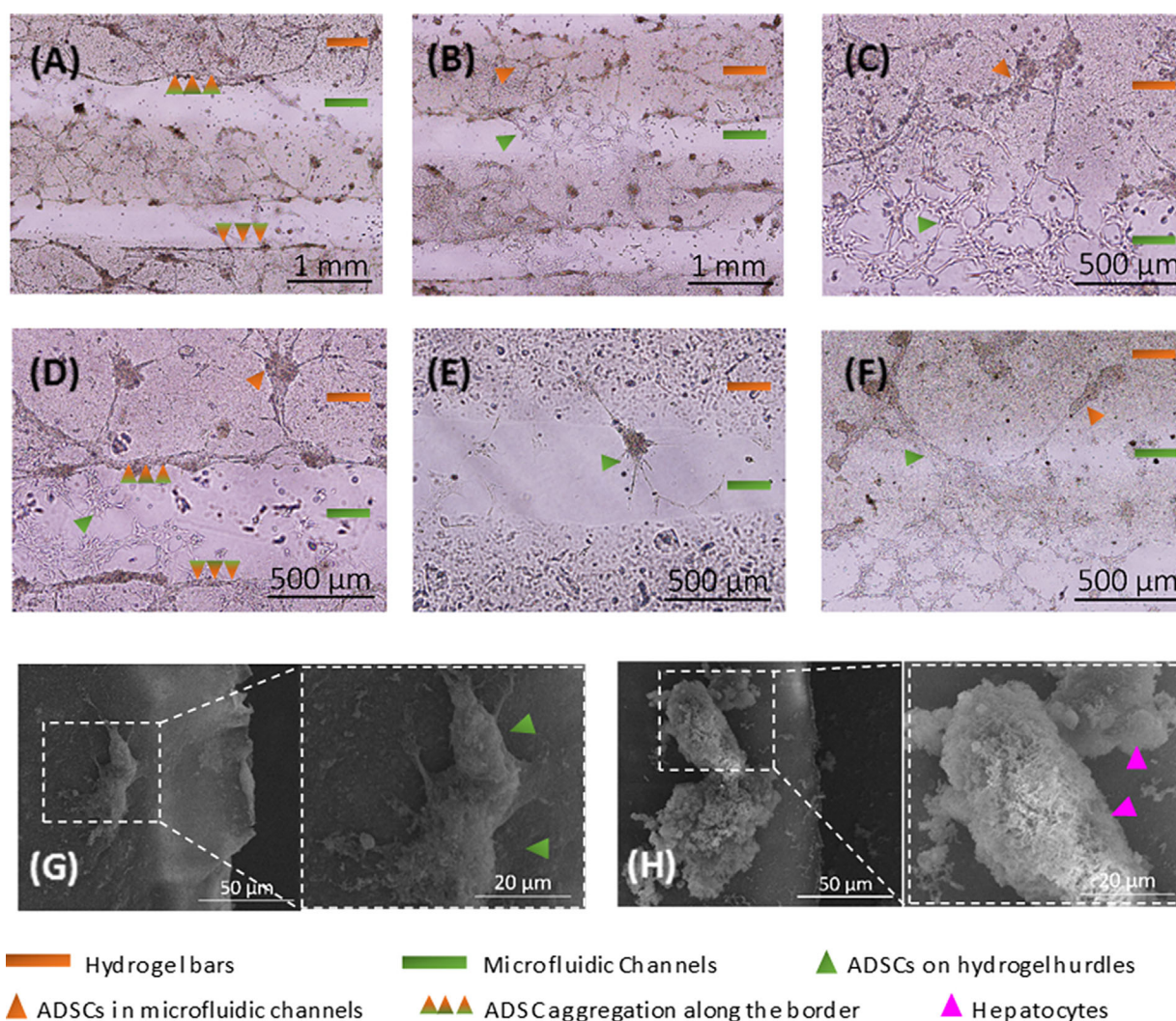


Figure 5. ADSC and hepatocyte morphologies along the microfluids between parallel hydrogel hurdles: (A) – (D) ADSC morphology after 10 h culture; (E) & (F) ADSC morphologies after 20 h culture; (G) SEM image of ADSCs on the edge of hydrogel hurdles; (H) SEM image of hepatocytes on the edges of the alginate/gelatin/fibrin hurdles.

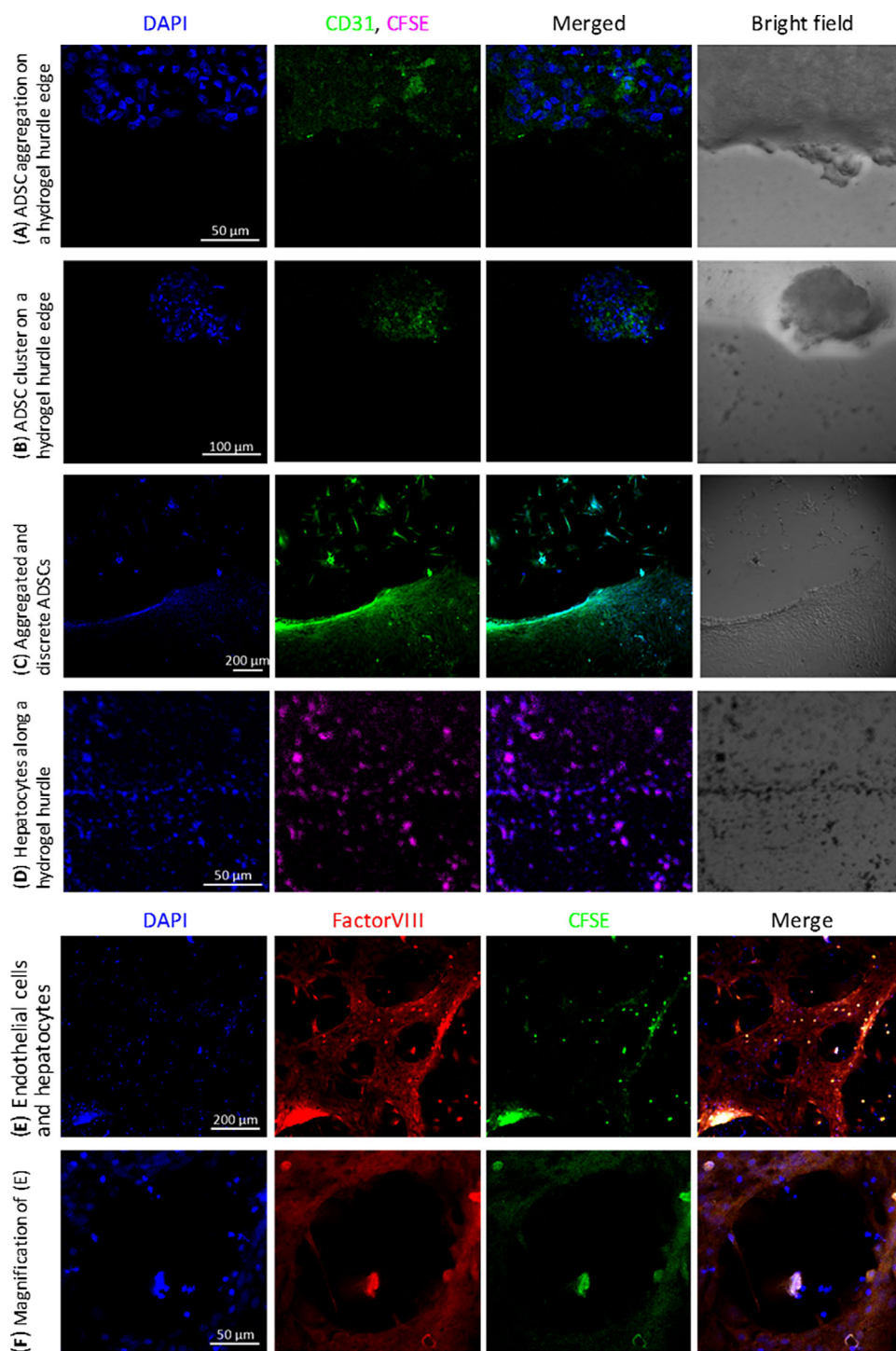


Figure 6. ADSC and hepatocyte morphologies with immunostaining: (A) & (B) ADSC aggregations and clusters in the alginate/gelatin/fibrin hurdles after 20 h culture; (C) ADSC proliferation in the microfluidic channels after 20 h culture; (D) hepatocyte distribution on the construct; (E) vascularization of ADSCs after co-cultured with hepatocytes; (F) magnification of (E).

phenomena in Figure 5A and D. In nearly all the microfluidic channels, ADSCs gathered in large population (Fig. 6C). Most of the ADSCs formed round spheroids with tight cell-cell connections and smooth surfaces on or in the hydrogel hurdles (Fig. 6A and B; 3D environment), while ADSCs in the micro channels tended to exhibit

a flat and elongated morphology (Fig. 6C; 2D environment). Figure 6D shows the even distribution of hepatocytes in the Q3D construct. No obvious hepatocyte clusters were found. Some hepatocytes tended to scatter along the hydrogel hurdles (Fig. 6D). In addition, the hepatocytes were more likely to attach to the

hydrogel hurdles and micro channels, possibly because of the large amount of protein secretion for liver functionalities. Endothelialization phenomena were found with this lab-chip technique after co-culture of the ADSCs and hepatocytes in the flowing DMEM. After 20 h of co-culture, plentiful Factor VIII expression was observed with the ADSCs (Fig. 6E and F). More importantly, the ADSCs were assembled into lumens and tubes, showing the formation of vascular capillaries. In contrast, most of the hepatocytes were still scattered with the presence of the endothelial-like cells, either in the center of lumina or along the cavity walls.

Computer Simulation

Figure 7 shows the fluid velocity in the alginate/gelatin/fibrin micro channels with four different channel widths (i.e., 4 mm, 2 mm, 1 mm, and 0.5 mm.). The cylindrical inlets of the flow were all upright with an inlet velocity of 3 mm/s. The disorders of fast flow at the inlets and outlets were found. The size of turbulence increased with decreasing channel width (Fig. 7A–D). Apart from the disorders in certain areas, all the fluids inside the micro channels remained steady and placid with a uniform velocity near 0 mm/s. However, the four graphs were generally analogous.

Discussion

Recently, microfluidic and 3DP technologies have shown some revolutionary impacts on tissue engineering, organ manufacturing and biological analysis (Figeys and Pinto, 2000). The integration of microfluidic movements, cell manipulations, and RP techniques is becoming increasingly important for regenerative medicine, high throughput drug screening, cellular toxicity testing, and biofluid analyses, since the optimization of biomaterial components, the engagement of stem cells to different cell lineages, and the controlled release of growth factors are key factors for tissue engineering and organ manufacturing (Cushing et al., 2008; Lee et al., 2008; 2012, 2013; Wang and Zhang, 2011; Wang et al., 2007, 2009, 2010, 2013). Some bioactive and compatible polymers have been used frequently in 3DP techniques to create biomimetic constructs with various levels of complexity. Some of the micro channels constructed with soft hydrogels have been applied for blood vessel formation (Resnick et al., 2003; Risau, 1997; Yamamoto et al., 2003). Especially, synthetic polymers have been employed as mechanical supports for their natural counterparts (Duffy et al., 1998; Huang et al., 2013; Hwang et al., 2013; Kaigala et al., 2007; Liu et al., 2011; Wang et al., 2013; 2015). Natural polymers overweigh the synthetic ones in terms of biocompatibility, biotoxicity and

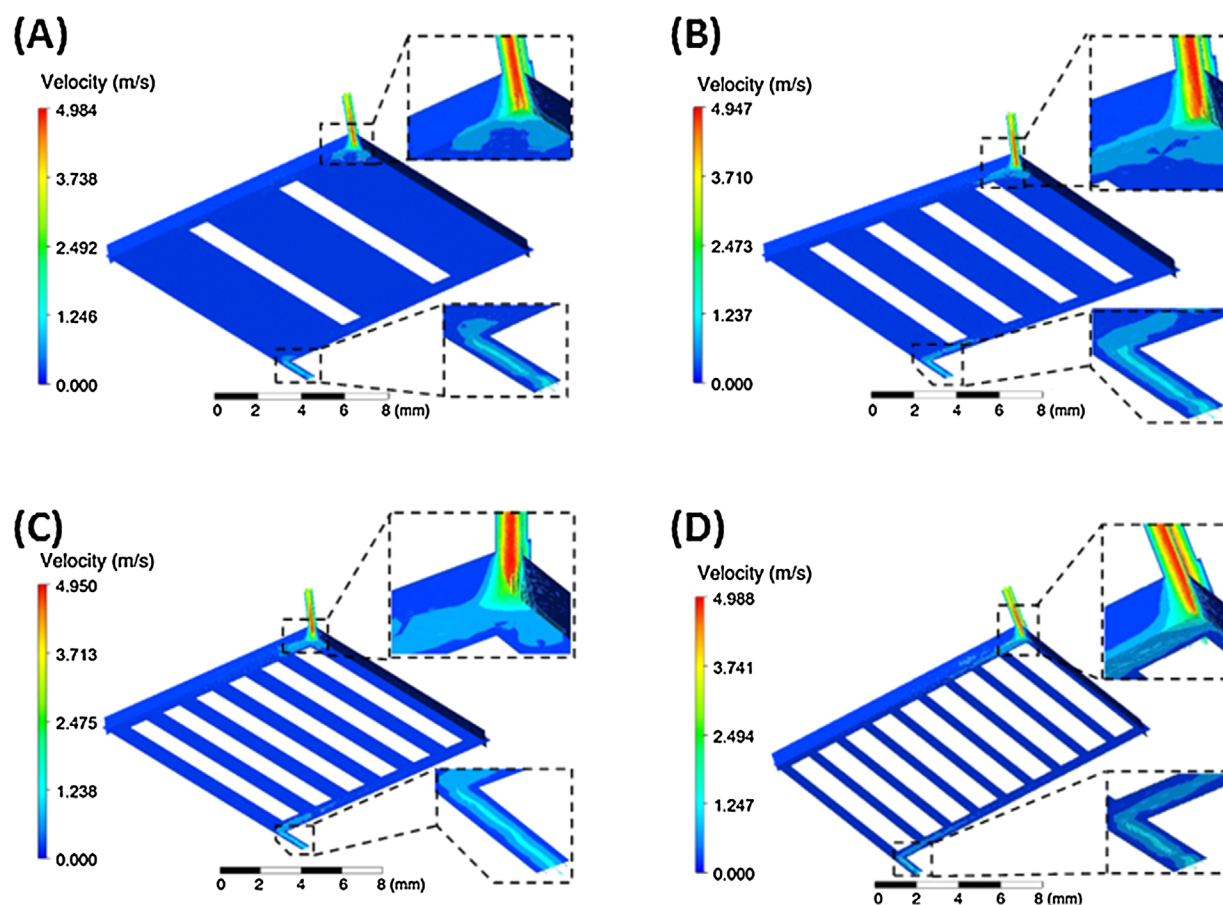


Figure 7. Simulation of fluid velocity in micro channels with different widths: (A) 4 mm; (B) 2 mm; (C) 1 mm; (D) 0.5 mm. The inlet of the fluid is cylindrical with a diameter of 300 μm , and the width of the outlet is 500 μm . The length of the channel is 13 mm. Fluidic analyses were carried out with ANSYS 15.0 (ANSYS).

degradability in vivo and/or in vitro despite their defects in mechanical performance (Liu et al., 2015; Xu et al., 2009). Among the frequently used biomaterials, collagen, alginate, chitosan, fibrin, gelatin and their composites are ideal ECMs for cell encapsulation and drug delivery.

In the present study, a Q3D construct was fabricated using parallel biocompatible gelatin/alginate/fibrinogen hydrogel with microfluidic channels (Fig. 2). CaCl_2 and thrombin solutions were used to crosslink/polymerize the alginate and fibrinogen molecules, respectively. With the distinct boundary structure of the gelatin/alginate/fibrin hurdles, both computer simulation and optical observation of the behaviors of the microfluidic flows and cells were possible (Fig. 2C; Fig. 5; Fig. 7). The microfluidic channels between the gelatin/alginate/fibrin hurdles formed passageways for various fluidic flows, either containing fluorochromes or cells (Fig. 2D and E; Fig. 3A and B; Fig. 4).

With the LSCM, the distributions of the fluid flows with fluorochromes were observed at first. The fluorescence intensity along the gelatin/alginate/fibrin hydrogel hurdles was reflected automatically. Each fluorochrome had its own exchange rate in the fluid, and the diffusion coefficients of different fluorochromes varied greatly. At the beginning of the experiment, the distribution areas of the fluorescein and rhodamine B were remarkably different: most of the fluorescein scattered in the middle of the micro channel, while vast of the rhodamine B concentrated on the border of the gelatin/alginate/fibrin hurdles (Fig. 3A and B). There were two sharp peaks in the intensity curve of the rhodamine B (Fig. 3A), while there were no remarkable intensity fluctuations in that of the fluorescein (Fig. 3B). When the two fluorochrome solutions jointed, both fluorescein and rhodamine B distributions became homogeneous and alike, which represented the uniform mixing of the two liquid fluids (Fig. 3C and D). The enrichment of rhodamine B on the boundaries of the gelatin/alginate/fibrin hydrogels (Fig. 3A) may be explained by substance adsorption and capillarity effect. It is therefore concluded that the mutual dissolution characteristics of the two fluorochromes were mainly influenced by the fluid properties such as the solution concentration and viscosity.

Figure 2F and Figure 4 show the cell-loaded fluid behaviors in the micro channels. In Figure 2F ADSCs and hepatocytes suspended in the micro channels were abreast to each other under a phase contrast microscope over the gelatin/alginate/fibrin hurdles. Nuclei of the ADSCs and hepatocytes were visualized dynamically under the LSCM (Fig. 4A and B). The color-coded time-lapse image of the ADSC nuclei was composed of 10 monochromatic frames with the interval time of 500 ms (Fig. 4C). It revealed that most of the cells moved straightly with the liquid flow along the channels, while a few cells shifted in the direction against the fluid flow. The fluidic turbulences of the liquids in the channels were quite complex, and the turbulences overweighed laminar flows under such circumstances. In addition, some ADSC aggregations stood still in the flows. This may be explained by assuming that the ADSCs had already attached to the bottom of the micro channels, the friction was strong enough to prevent the cells from moving forward, or the impact force of the liquid was insufficient to engine cells to move forward.

The morphology of cell types altered over time in the Q3D constructs. On the one hand, most of the ADSCs in the micro

channels formed networks connecting the neighbor gelatin/alginate/fibrin hurdles after 10 h of culture (Fig. 5B and C). The scale of the networks became larger and larger from then, penetrating and traversing several remote micro channels after 20 h (Fig. 5F). The star-shaped ADSC aggregates had plenty filamentary fibers. On the other hand, some of the ADSCs gathered along the edge of the hydrogel hurdles and formed linearly arrays and heavy aggregations (Fig. 5A and 5D). Some of the ADSCs proliferated and aggregated on the coverslips and hydrogel hurdles. The reason for the ADSC aggregation and growth along the borders of the hydrogel hurdles may be due to the special topological structure, mechanics support, nutrients enrichment, and/or comprehensive physico-chemical effects. Nutrients in the DMEM were likely to be absorbed by the gelatin/alginate/fibrin hydrogels and cells on the boundaries as well, in accord with the rhodamine B enrichment effects in Figure 3A. The flow shear stress and hydrogel thickness may also contribute to the growth of ADSCs. Even the uninterrupted cell lines and clusters may play some roles in the stabilization of the gelatin/alginate/fibrin hydrogel structure for a while. With the coverage of the cell lines, the 3D printed hydrogel hurdles were further solidified. In contrast, the distribution of the spherical hepatocytes was totally different with the ADSCs with no sign of elongated pseudopodia (Fig. 5H). The proliferation rate of the hepatocytes was very low in the microfluidic channels. In addition, the hepatocyte aggregations became smaller and smaller during the later experimental stages. This result is contrary to the other studies, in which hepatocytes could form multicellular clusters and spheroids in some hydrogels (Glicklis et al., 2000). The reason may be that the growth environments of hepatocytes on the hydrogel surfaces were totally different from those in the hydrogels, and the fluidic flow field may prevent the hepatocytes to further aggregate.

For CFD simulation, different widths of the channels (i.e., 4 mm, 2 mm, 1 mm, and 0.5 mm) were applied to the stereo models (Fig. 2; Fig. 7). The results of the four groups were similar. The disorders of high flow velocity (approximately 3 mm/s) were mainly found at the inlets and outlets. However, the fluid in the microfluidic channels stayed steady and almost still with a uniform velocity near 0 mm/s. Besides, the region of disorder enlarged with decreasing channel width (Fig. 7A to 7D). The speedy liquid columns may lash against the fluid inlets, but the flow in the channels remained almost undisturbed, which is consistent with the preservation and stabilization of the gelatin/alginate/fibrin hydrogels (Fig. 2C; Fig. 5). Though CFD can simulate and analyze fluid dynamic movements with high efficiency, cost saving and dispersed numerical value, and can simulate physical analogs such as non-steady flow, laminar flow, turbulent flow, compressible flow and incompressible flow, the merits and defects of computational dynamics should be meanwhile considered. There are still some gaps between the lab results and computational simulations, especially for the accuracy and fidelity in biological and fluidic procedures in micro- and nano-scale [Zheng et al., 2011].

Recently, innovative cell-based therapies offer promises for the repair and regeneration of tissues or organs. Embryonic stem cells, induced pluripotent stem cells and adult stem cells were frequently used for vascular tissue engineering (Fernandes et al., 2010; Wagers, 2012; Yang et al., 2014). Among these stem cells, ADSCs are an ideal cell source because they are easily accessible in abundant

numbers and can differentiate along multiple lineage pathways (Gimble et al., 2007). Previous studies documented the ability of ADSCs to undergo endothelialization in vitro with largely expressed CD31 and FactorVIII (Jaffe, 1977; Lin et al., 2010; Sehested and Hou-Jensen, 1981). ADSCs have shown the ability to form vascular networks with the de novo growth of vascular capillaries (Fraser et al., 2006; Planat-Benard et al., 2004; Sengenès et al., 2005). Since the genesis of blood vessels or vascular systems is crucial for organ development and repair. Techniques involved in endothelial cell generation and self-assembly are emerging (Davis and Senger, 2005; Richardson et al., 2001; Stratman et al., 2009). Some efforts have focused on the interactions of vascular seed cells and natural polymers (Hughes, 2008; Jeong et al., 2011; Wang and Wang, 2015; Zhao and Wang, 2013). Although the fine adjustment of angiogenesis and vascularization in vivo has been well-studied, the vascularization guided by chemicals and biomaterials in vitro still remains to be unsolved (Kamei et al., 2006).

In the present study, we also examined the ADSC and hepatocyte mixtures using immune staining under the LSCM. Cellular morphological changes and differentiation were demonstrated in the microfluidic channels. Even under the microfluidic flow, ADSCs tended to gather along the boundaries of the hydrogel hurdles with round spheroids and compact intercellular connections (Fig. 6A and B). In the microfluidic channels, ADSCs aggregated in a flat and elongated morphology (Fig. 6C). It could be concluded that the biocompatible gelatin/alginate/fibrin hydrogel hurdles can be used for both 2D and 3D physical environment imitation, where the 2D milieu was mainly along the microfluidic channels and 3D condition was created on or in hydrogel matrix. Hepatocytes were dispersively located in the Q3D construct, with no signs of aggregations but plenty secretions (Fig. 6D). The protein secretion ability of the hepatocytes was maintained, which holds potential for drug screening and toxicological testing for livers. The co-culture of ADSCs and hepatocytes promoted the endothelialization of ADSCs with repetitive capillary networks marked with FactorVIII (Fig. 6E and F). These findings suggested that hepatocytes may release vascularization-related growth factors such as VEGF, TGF β and HGF, creating a naturally biological cocktail for the development and maturation of de novo vascular systems. The interactions among homogeneous and heterogeneous cells indicated histogenesis and endothelialization along with stem cell differentiation and proliferation (Fig. 6E and F). Thus, this integration represents a new pathway for the studies of intercellular and cell-material interactions. The co-culture of stem cells and somatic cells attributes to the differentiation of the stem cells into the related phenotypes (Soto-Gutiérrez et al., 2007). With this protocol, cell-cell communications or interactions were guaranteed (Kloxin et al., 2010). Further research on the co-culture effect of stem/progenitor cells and adult cells are expected in both 2D and 3D microfluidic systems for the basic investigation on organ manufacturing and regenerative medicine. In the future, large-scale integration of microfluidics and open-source 3DP processing apparatus may trigger the abundance and versatility of analysis methods and biomedical treatments, offering sufficient biomimetic and delicate resources for controlled chemical release, drug screening, neuroscience, migration and chemotaxis, tissue models and cell mechanics, cell manipulation and genetic research in pharma-

ceutical industry (Franzesi et al., 2006; Pearce, 2012; Thorsen et al., 2002; Zhao et al., 2014). Specifically, the signaling pathways and regulating factors in drug-controlled-release systems should be optimized spatial-temporally. Further integrations of RP and fluidic techniques would promise a much more ideal pathway for multiple cell organizations, zonal bio-chemical reactions, and dynamic cell-biomaterial interactions.

Conclusions

A Q3D construct with parallel biocompatible alginate/gelatin/fibrin hurdles was printed for microfluidic and cell mobility analyses. Fluid and cell behavior changes were found with microfluidic flows containing fluorescein, rhodamine B, ADSCs, and hepatocytes. Microfluid velocities in the micro channels with various widths were matched with those of the CFD results. Different fluorochrome solutions and cells behaved extraordinary different in the micro channels. Morphological changes of cells, especially ADSCs in the 2D and 3D milieu revealed some unusual patterns. Especially, co-culture of ADSCs and hepatocytes demonstrated an alternative approach for in vitro endothelialization and vascularization. These results highlight the huge possibility for the manipulation of cells and fluids at micro-scale integrated with RP techniques. This approach therefore could bridge the gaps between the RP engineering methods and microfluidic biomedical approaches.

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